To calculate the random forces associated to the kth term of the expansion, we note that these can be viewed as the solution of a linear SDE, specifically

$$\mathrm{dR}_{i,k} = -\frac{1}{\tau_k}\mathrm{R}_{i,k}(t)\mathrm{d}t + \frac{1}{\tau_k}\sqrt{2\mathrm{k_B T}c_k}\mathrm{dW}_{i,k}(t) \tag{8.65}$$

where $\mathrm{W}_{i,k}$ is a standard Wiener process. The random force is seen to be defined as the solution of an Ornstein-Uhlenbeck process induced by the Prony representation. It is easily shown that $\mathrm{R}_{i,k}(t)$ has zero mean and autocorrelation defined by

$$\langle \mathrm{R}_{i,k}(t+s)\mathrm{R}_{i,k}(t)\rangle = \mathrm{k}_B\mathrm{T}\frac{c_k}{\tau_k}\exp\left[-\frac{1}{\tau_k}s\right], \quad s \geq 0 \tag{8.66}$$

The total random force is then obtained by summation

$$\mathrm{R}_i = \sum_{k=1}^{k_{\max}} \mathrm{R}_{i,k}. \tag{8.67}$$

Observe that both random and drag forces are incorporated via summation of the components $\eta_{i,k}$ and $\mathrm{R}_{i,k}$ defined by linear equations, respectively the ODE (8.64) and SDE (8.65). It is therefore possible to replace these components by their sum

$$\sigma_{i,k} = \eta_{i,k} + \mathrm{R}_{i,k}$$

which may be evolved by a single SDE. The equations therefore become

$$\mathrm{d}q_i = m_i^{-1}p_i\mathrm{d}t, \tag{8.68}$$

$$\mathrm{d}p_i = F_i\mathrm{d}t + \sum_{k=1}^{k_{\max}}\sigma_{i,k}\mathrm{d}t, \tag{8.69}$$

$$\mathrm{d}\sigma_{i,k} = -\frac{1}{\tau_k}\sigma_{i,k}\mathrm{d}t - \frac{c_k}{\tau_k}p_i\mathrm{d}t + \frac{1}{\tau_k}\sqrt{2\mathrm{k_B T}c_k}\mathrm{dW}_{i,k}. \tag{8.70}$$

The relation between the Prony series representation and Langevin dynamics can be seen in an informal sense by multiplying the equation for $\sigma_{i,k}$ equation by τ_k, and considering $\tau_k \to 0$, giving

$$\sigma_{i,k}\mathrm{d}t = -c_k p_i\mathrm{d}t + \sqrt{2\mathrm{k_B T}c_k}\mathrm{dW}_{i,k},$$

which results in

$$\mathrm{d}p_i = F_i\mathrm{d}t - \left(\sum_{k=1}^{k_{\max}} c_k\right)p_i\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\sum_{k=1}^{\ell}\sqrt{c_k}\mathrm{dW}_{i,k}.$$

Observing that the sum of ℓ independent Guassian random variables of variance c_k is a Gaussian random variable with variance $\sum_{k=1}^{\ell} c_k$, we see that this is in the form of Langevin dynamics

$$\mathrm{d}p_i = F_i \mathrm{d}t - \hat{\gamma} p_i \mathrm{d}t + \sqrt{2\mathrm{k_B T}\hat{\gamma}}\mathrm{dW}_{i,k},$$

with

$$\hat{\gamma} = \sum_{k=1}^{N_k} c_k.$$

(For the formal treatment of this limit refer to [293].)

For numerical integration, the system (8.68)–(8.70) can be handled via splitting. A natural family of methods based on Verlet is the following [19]:

Generalized Langevin Dynamics Method

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &:= \boldsymbol{p}_n + \frac{h}{2}\boldsymbol{F}(\boldsymbol{q}_n) + \frac{h}{2}\sum_{k=1}^{k_{\max}} \boldsymbol{\sigma}_{n,k}, \\
\boldsymbol{q}_{n+1} &:= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}, \\
\boldsymbol{\sigma}_{n+1,k} &:= \theta_k \boldsymbol{\sigma}_{n,k} - (1-\theta_k)\, c_k \boldsymbol{p}_{n+1/2} + \alpha_k \sqrt{2\mathrm{k_B T} c_k}\, \boldsymbol{Z}_k^n, \\
\boldsymbol{p}_{n+1} &:= \boldsymbol{p}_{n+1/2} + \frac{h}{2}\boldsymbol{F}(\boldsymbol{q}_{n+1}) + \frac{h}{2}\sum_{k=1}^{k_{\max}} \boldsymbol{\sigma}_{n+1,k},
\end{aligned}$$

where $\boldsymbol{\sigma}_{n,k}$ is a $\mathrm{N_c}$-dimensional vector, and the components of the $\mathrm{N_c}$-dimensional vectors $\mathbf{Z}_{n,k}$ are independent standard normal random variables.

The real-valued coefficients θ_k and α_k which must satisfy, for consistency, the equations

$$\theta_k = 1 - \frac{h}{\tau_k} + \mathcal{O}(h^2), \qquad \alpha_k = \frac{\sqrt{h}}{\tau_k} + \mathcal{O}(h).$$

An argument is given in [19] that α_k and θ_k should be taken as

$$\theta_k = \exp(-h/\tau_k), \qquad \alpha_k = \sqrt{\frac{(1-\theta_k)^2}{h}},$$

since, for this choice, the Langevin limit is preserved. Alternative methods are possible. For many comparisons and numerical experiments, refer to [19].